1. Functional Requirements

Table 1.1: Functional requirements

ID	Function	Description
FR1	Ingestion	Import STEP/BREP files headlessly; extract the assembly tree, transforms, mass properties, holes and planes.
FR2	Geometric identity	Compute a pose-invariant fingerprint per solid; a re-exported, renamed or rotated solid maps to the existing shape record.
FR3	Assembly hashing	Maintain a Merkle root hash so two configurations compare in constant time and differences are located by descent.
FR4	Contracts	Store each revision's public interface: envelope, hole patterns, mass limit, CG window, datums and attributes.
FR5	Commit validation	Validate a change as a unit, classify it as no-op, internal or breaking, and reject violations naming the constraint and team.
FR6	Concurrency	Accept concurrent commits without locking, using compare-and-swap on the root hash.
FR7	Tolerance chains	Evaluate declared chains by worst-case and RSS stacking; flag chains that cross team boundaries.
FR8	Structural queries	Answer BOM, mass rollup, where-used, impact, substitution and configuration-difference queries.
FR9	History	Keep an append-only commit log with a full validation trace for every commit.
FR10	Export	Rebuild any part or assembly as a STEP file for viewing in CAD tools such as FreeCAD.
FR11	User interfaces	Provide a command-line tool, a browser-based review interface and optional natural-language queries.

2. Non-Functional Requirements

- **Performance:** structural queries return in milliseconds and never invoke the geometry kernel.
- **Correctness:** duplicate shapes are impossible by construction; a 0.2 mm hole change yields a new fingerprint.
- **Integrity:** commits are all-or-nothing, revisions are never edited in place, and assembly cycles are rejected.
- **Storage efficiency:** storage grows with distinct geometry, not with the number of revisions.
- **Usability:** every rejection names the violated constraint, the affected part and the owning team.
- **Portability:** runs offline on Windows, Linux and macOS; the natural-language layer is optional.
- **Maintainability:** modular packages with an automated pytest suite.

3. Hardware Requirements

Table 3.1: Hardware requirements

Component	Minimum	Recommended
Processor	64-bit dual-core, 2.0 GHz	Quad-core, 3.0 GHz or higher
Memory (RAM)	4 GB	8 GB or more
Storage	2 GB free	10 GB free on an SSD
Display	Not required (kernel runs headless)	1920 × 1080 for the review interface
Network	Not required (runs offline)	Internet only for the optional NL layer

4. Software Requirements

Table 4.1: Software requirements

Software	Version	Purpose
Operating system	Windows 10/11, Ubuntu 22.04+, macOS 12+	64-bit host platform
Python	3.10	Implementation language
SQLite	Bundled with Python	Relational store, recursive queries
cadquery-ocp (Open CASCADE)	7.9.3.1.1	Geometry kernel, used only at import
NumPy, SciPy	2.2.6, 1.15.3	Numerical computation, socket matching
TinyDB	4.9.0	Document store for attributes
FastAPI, Uvicorn, Jinja2	0.136.3, 0.32.0, 3.1.6	Browser-based review interface
anthropic, python-dotenv	1.7.0, 1.0.1	Optional natural-language layer
pytest, httpx	9.0.3, 0.28.1	Automated testing
FreeCAD (optional)	1.1.3	Viewing exported STEP files